

Assignment

Course Code: 0613-205

Course Name: Algorithms

1. What is recursion, and how does it differ from iteration?
2. Define base case in recursion. Why is it important?
3. Explain direct recursion and indirect recursion with examples.
4. Write a recursive function to calculate the factorial of a number.
5. What are the drawbacks of recursion in terms of performance and memory?
6. When should recursion be avoided in algorithm design?
7. Define the level of a tree with an example.
8. Explain the height of a tree with an example.
9. Application of a tree.
10. Define the degree of a tree with an example.
11. Compare binary tree and binary search tree.
12. Write some advantages and disadvantages of binary search trees.
13. Define a balanced binary search tree.
14. Dijkstra's algorithm for finding the shortest path from source to destination.

